$$\mathsf{check}^{k}_{j}(\mathrm{tt}, v) \dashrightarrow v \qquad \text{Check-True}$$

$$\mathsf{check}^{k}_{j}(\mathrm{ff}, v) \dashrightarrow \mathsf{blame}^{k}_{j} \qquad \text{Check-False}$$

$$\mathsf{mon}^{k,l}_{j}(\mathsf{flat}(e), v) \dashrightarrow \mathsf{check}^{k}_{j}(e\ v, v) \qquad \text{Mon-Flat}$$

$$\mathsf{mon}^{k,l}_{j}(\kappa, v) \dashrightarrow \mathsf{guard}^{k,l}_{j}(\kappa, v) \qquad \text{Mon-Guard}$$

$$\text{where } \kappa \neq \mathsf{flat}(e)$$

$$\mathsf{guard}^{k,l}_{j}(\kappa_1 \to \kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}^{k,l}_{j}\Big(\kappa_2, v_1\ \mathsf{mon}^{l,k}_{j}(\kappa_1, v_2)\Big) \qquad \text{Guard-Func}$$

$$\mathsf{guard}^{k,l}_{j}(\kappa_1 \to_i \lambda x.\kappa_2, v_1)\ v_2 \dashrightarrow \mathsf{mon}^{k,l}_{j}\Big(\kappa_3, v_1\ \mathsf{mon}^{l,k}_{j}(\kappa_1, v_2)\Big) \qquad \text{Guard-Dep}$$

$$\text{where } \kappa_3 = \Big\{\mathsf{mon}^{l,j}_{j}(\kappa_1, v_2)/x\Big\}\kappa_2$$

$$\mathsf{get}\Big(\mathsf{guard}^{k,l}_{j}(\kappa, v)\Big) \dashrightarrow \mathsf{mon}^{k,l}_{j}(\kappa, \mathsf{get}(v)) \qquad \text{Mon-Get}$$

$$\mathsf{set}\Big(\mathsf{guard}^{k,l}_{j}(\kappa, v_1), v_2\Big) \dashrightarrow \mathsf{mon}^{k,l}_{j}(\mathsf{ref/c}(\kappa), e) \qquad \text{Mon-Set}$$

$$\text{where } e = \mathsf{set}\Big(v_1, \mathsf{mon}^{l,k}_{j}(\kappa, v_2)\Big)$$

$$\frac{E \neq \square}{E\big[\mathsf{blame}^{k}_{j}\big], \sigma \to \mathsf{blame}^{k}_{j}, \sigma}\ \text{Step-C-Blame} \qquad \frac{e_1 \dashrightarrow e_2}{E[e_1], \sigma \to E[e_2], \sigma}\ \text{Step-C}$$

Figure 1: Reduction Rules for Contracts